Fahad Mollah

AI Engineering Student | Applied AI, Data & Autonomous Systems

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EXPERIENCE

Technohaven Company Limited, *Project Engineer* 06.2022 – 07.2024 | Dhaka, Bangladesh

- Worked on site at the Bangladesh Export Processing Zones Authority (BEPZA) on its Process Automation Project
- Helped 180+ non-technical staff move from paper-based work onto the new digital systems
- Sat with users to find out what was really blocking them, then fixed it or passed it to the right person
- Made sure staff followed computer use, information security and privacy policies on site
- Checked that the agreed security standards were in place on every system we delivered
- Handled server and network support for the project and solved day-to-day technical problems
- Checked the quality of data going into the new systems and prepared regular reports for the project team

PROJECTS

docuagent: Document Question Answering Tool, 2026 – present

Personal Project | Python, LangChain, ChromaDB, Groq LLM API

- Built a tool that answers questions about PDF documents and cites the source passage
- Read PDFs with PyMuPDF, split them into overlapping chunks and stored the embeddings in ChromaDB
- Prompted the LLM to answer only from the retrieved passages and say when the answer is missing
- Adding pytest unit tests so changes can be checked automatically (in progress)

Reinforcement Learning Training Pipeline for Autonomous Inspection, 03.2025 – 07.2025

THI Project | Python, Pygame, Q-learning, DQN

- Built a custom simulation environment and trained Q-learning and DQN agents from scratch
- Used experience replay, reward shaping and epsilon-greedy decay to make learning more efficient
- Set up a reproducible pipeline with fixed seeds and structured configs to compare agent variants fairly
- Tracked success rate and convergence across runs; tuning raised success rate from 50% to 95%

Autonomous Object Sorting System, 03.2025 – 07.2025

THI Project | C++, Arduino, Braccio Arm, OpenCV

- Programmed pick-and-place routines for a Braccio robotic arm
- Added an OpenCV colour detection module so the arm could classify and sort objects on its own
- Tested the final version over 15 runs, correct object sorted in 14 of 15 (93%)

EDUCATION

Technische Hochschule Ingolstadt (THI), 03.2024 – present | Ingolstadt, Germany

Master of Engineering in AI Engineering of Autonomous Systems

- Focus: machine learning, computer vision and robotics
- Reinforcement learning (Q-learning, DQN) applied hands-on in course projects

American International University-Bangladesh (AIUB), 01.2018 – 06.2022 | Dhaka, Bangladesh

Bachelor of Science in Software Engineering

SKILLS

Programming: Python, C++, SQL, JavaScript

ML & Data: PyTorch, TensorFlow (basic), OpenCV, pandas, NumPy, scikit-learn, Feature Engineering, EDA

AI Foundations: LLM and Transformer Architectures, Retrieval-Augmented Generation (RAG), Agent-Based AI Systems

Autonomous Systems: Sensor Data Fusion (Camera, Radar, LiDAR), System Identification, Simulation, Edge Computing

Tools: Git, Docker (basic), Jupyter, Linux/CLI, VS Code, MS Excel

LANGUAGE

English (C1), Deutsch (A1), Bangla (Native)